

Reference excerpt 02-003

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annihilate at one blow ! CONDITIONS OF USE This is a self-published methodological note distributed under the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The note contains an original reasoning of mine and the goal to share thoughts and methodologies, not results. Therefore before using the contents of these notes, everyone is invited to verify the accuracy of the assumptions and conclusions. INTRODUCTION Jugular venous pulse (JVP) examination dates back to more than a century ago [Mackenzie(1902), Mackay(1947), Applefeld(1990), Kalmanson et al. (1972)], but only recently it has been shown that the trace of the JVP can be obtained by means of an ultrasound scanner (US) equipped with a commercial linear probe.[Sisini et al.(2015), Sahani et al.(2015)] The method proposed needs a video clip of a few seconds obtained from the transverse scan of the neck, where the internal jugular vein (IJV) is clearly visible. [Sisini et al. (2015)] The cross sectional area of the IJV (CSA) is measured on each picture (sonogram) of the video clip. The sequence of measurements (data sets) is the CSA trace. Moreover, in the same studies, it was shown that acquiring the video clip of the IJV simultaneously to the ECG trace, allows to identify relationship between the two ECG and JVP traces (see Fig. 1). The non-invasive techniques used up to now, are based on the use of a microphone or a motion sensor and produce a trace